Dr. Madangopal Krishnan
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सत्यमेव जयते

Government of India
भारतसरकार
Bhabha Atomic Research Centre
भाभापरमाणुअनुसंधानकेन्द्र
Fusion Reactor Materials & Laser Processing Section
Materials Science Division

Ref: BARC/MSD/(Trg.Cert)/2019/131940

July 11, 2019

CERTIFICATE

This is to certify that **Shri. KALIDASS M, 3rd year B.E Department of Metallurgical Engineering, PSG College of Technology, Coimbatore, Tamil Nadu, India,** has undergone summer training in **Materials Science Division, Bhabha Atomic Research Centre**, Trombay, Mumbai-400 085 for the period from **May 13, 2019 to July 11, 2019** under the guidance of **Dr. S.P. Chakraborty, Head, Fusion Reactor Materials & Laser Processing Section, Materials Science Division.**

During the above period, he was involved in the following project works entitled 1. **Electron Beam and Laser Hybrid welding Studies on Ni-Cr-Mo based Super alloy of C-276 variety for optimization of Process parameters, 2. Excimer Laser ablation cleaning of Mo based TZM alloy, Cu & Cu based brass alloy.** During his tenure, Shri. Kalidass M has actively participated in the welding studies on Ni-Cr-Mo based Super alloy, namely, Hastelloy C276 by Electron Beam and Laser-TIG hybrid welding. Bead-on plate and butt joint welds were characterized by Micro-hardness measurement and Optical microscopy. There is no significant heat affected zone due to high voltage and low current in the EB welding process used. The hardness of the weld zone is slightly higher than base metal. Defect free weld beads with good penetration were obtained at **250 J/mm heat input** and **10 mm/sec welding speed.** A fine lamellar structure, without defects is achieved in the weld zone by electron beam welding. Hence it can be used as-weld condition in applications. Excimer Laser ablation was done to clean the tarnished surfaces of Mo based TZM alloy, Cu and Cu based Brass alloy. These laser ablated cleaned and uncleaned surfaces were examined by Optical Microscopy. Micro-hardness and roughness were measured by micro-hardness tester and 3D profilometer. The results demonstrated sufficient cleaning of the surfaces in nano-meter ranges without affecting the base materials.

During the period of his training, he was found to be sincere, hardworking with excellent conduct and work performance.

I extend my best wishes to **Shri. Kalidass M** for his bright future and success.

(Dr. Madangopal Krishnan)
डॉ. मदनगोपाल कृष्णन/ Dr. Madangopal Krishnan
सह निदेशक, पदार्थ वर्ग एवं
Associate Director, Materials Group &
अध्यक्ष, पदार्थ विज्ञान प्रभाग
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The Head,
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